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## Balanced repeated replications based on mixed orthogonal arrays

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### SUMMARY

For stratified samples with arbitrary numbers of primary sampling units per stratum, one can divide the units in the stratum into two groups and then apply the balanced half-samples method to the two groups thus formed. The efficiency loss of the grouped balanced half-samples variance estimator can be substantial. Use of a mixed orthogonal array, whose numbers of symbols correspond to the numbers of units per stratum, gives a set of balanced subsamples. Unlike that of Gupta & Nigam (1987), the proposed method of analysis gives a consistent variance estimator and bias reduction for nonlinear statistics. To further reduce the number of replicates, use of nearly orthogonal arrays is considered.

*Some key words:* Balanced half-samples; Grouping; Nearly orthogonal array; Sample quantile; Stratified sample.

### 1. INTRODUCTION

The balanced half-samples method (McCarthy, 1966, 1969) is commonly used for estimating the variance of nonlinear statistics from stratified samples with two primary sampling units per stratum. These samples arise naturally if deep stratification is desired as in some large-scale surveys. The method is extended (Gurney & Jewett, 1975) to the case of  $p$  primary sampling units per stratum for prime  $p$ . This extension for  $p > 2$  is not adopted in common practice since it requires a much larger number of replicates. Also, in many survey designs the numbers of primary sampling units are unequal. A common practice is to divide the units in the stratum into two groups and then to apply the balanced half-samples method to the two groups thus formed. This has gained popularity because the software for obtaining balanced half-samples is readily available and the number of replicates required is more economical. As shown in § 2, the resulting variance estimators can be quite inefficient and their bias based on a conditional analysis can be considerable. The conditional analysis also suggests a method of grouping to reduce the bias. The degree of grouping depends on the tradeoff between computational economy and statistical efficiency. If the efficiency loss from grouping is not justifiable, one may draw replicates by using mixed orthogonal arrays with variable numbers of symbols to accommodate the different numbers of units or groups of units in the strata. As long as an orthogonal array is used for drawing replicates, we call the method balanced repeated replications. In § 3 we show that the analysis method of Gupta & Nigam (1987) based on mixed orthogonal arrays leads to an inconsistent variance estimator for nonlinear statistics, and propose a method that is valid for variance estimation and bias reduction for linear statistics as well as nonlinear statistics that can be expressed as smooth functions of estimated totals. In § 4 we consider an extension of the proposed method for general

statistics that are smooth functionals of the sample distribution functions. To reduce the number of replicates, use of nearly orthogonal arrays may be considered. This and related issues are discussed in § 5.

## 2. EFFICIENCY LOSS OF THE BALANCED HALF-SAMPLES VARIANCE ESTIMATOR DUE TO GROUPING

In stratified random sampling  $n_h \geq 2$  units with values  $y_{hi}$ , for  $i = 1, \dots, n_h$ , are selected randomly with replacement from the  $h$ th stratum with population size  $N_h$ , for  $h = 1, \dots, L$ . An unbiased estimator of the population mean is  $\bar{y}_{st} = \sum W_h \bar{y}_h$ , where

$$\bar{y}_h = (y_{h1} + \dots + y_{hn_h})/n_h, \quad W_h = N_h/N, \quad N = \sum N_h.$$

An unbiased variance estimator for  $\bar{y}_{st}$  is

$$v(\bar{y}_{st}) = \sum_h P_h^2 n_h^{-1} \sum_i (y_{hi} - \bar{y}_h)^2, \quad (1)$$

where  $P_h^2 = W_h^2/(n_h - 1)$ .

When  $n_h = 2$  for all  $h$ , a balanced half-samples variance estimator for  $\bar{y}_{st}$  is (McCarthy, 1969)

$$v_b(\bar{y}_{st}) = R^{-1} \sum_r (\bar{y}_{st}^{(r)} - \bar{y}_{st})^2,$$

where  $\bar{y}_{st}^{(r)}$  is the mean of the  $r$ th half-sample  $y_{h\delta(r,h)}$ , for  $r = 1, \dots, R$ ,  $h = 1, \dots, L$ ,  $\delta(r, h) = 1$ , and respectively 2 if the  $(r, h)$ th entry of an  $R \times L$  submatrix of a Hadamard matrix is 1 and respectively  $-1$ . It is known that  $v_b(\bar{y}_{st}) = v(\bar{y}_{st})$ .

For general  $n_h$ , one can divide the  $n_h$  units in stratum  $h$  into two groups of units and then apply the  $v_B$  formula by treating the two groups as the two units in the balanced half-samples method. We call the resulting estimator the grouped balanced half-samples variance estimator and denote it by  $v_{GB}$ . If  $n_h = 2m_h$  and each of the two groups consists of  $m_h$  units,

$$v_{GB}(\bar{y}_{st}) = \frac{1}{2} \sum_{h=1}^L W_h^2 \sum_{i=1}^2 (z_{hi} - \bar{z}_h)^2, \quad (2)$$

where  $z_{h1}$  and  $z_{h2}$  are the averages of the  $m_h$  units in the first and second groups,  $\bar{z}_h = \frac{1}{2}(z_{h1} + z_{h2}) = \bar{y}_h$ . Kish & Frankel (1970) considered this method and showed that under random grouping  $E_r\{v_{GB}(\bar{y}_{st})\} = v_b(\bar{y}_{st}) = v(\bar{y}_{st})$ , where the subscript  $r$  denotes computations with respect to the random grouping. This unbiasedness result, which follows easily from (3) below, does not tell us the efficiency loss of  $v_{GB}$  due to grouping. Krewski (1978, form. (11)) showed that, under  $n_h = p = 2m_h$  and other restrictive conditions,

$$\frac{\text{var}\{v_{GB}(\bar{y}_{st})\}}{\text{var}\{v(\bar{y}_{st})\}} = p - 1,$$

where the variance is with respect to the random grouping and the sampling design. Although the precise result holds in a special case, it suggests that the efficiency loss can be quite severe even for  $p = 3$ . If  $n_h = p = t\lambda$  units are randomly divided into  $t$  groups of  $\lambda$  units, J. N. K. Rao showed in an unpublished note that under the same restrictive conditions

$$\frac{\text{var}\{v_{GB}(\bar{y}_{st})\}}{\text{var}\{v(\bar{y}_{st})\}} = \frac{p-1}{t-1}.$$

The efficiency loss is drastically reduced by increasing  $t$  from 2 to, say, 4 groups. In practice this reduction in efficiency loss should be weighed against the increased cost of computations. For example, if  $n_1 = 4$ ,  $n_h = 3$  for  $h = 2, \dots, 14$ ,  $n_h = 2$  for  $h = 15, \dots, 51$ , we can use the balanced half-samples method with 52 replicates by grouping the units in each of the first 14 strata into two units. The 52 replicates are chosen according to the rows of a  $52 \times 51$  Hadamard matrix. If the efficiency loss due to grouping is severe, we can modestly increase the replicates to 72 and use a mixed orthogonal array OA  $(72, 4 \times 3^{13} \times 2^{37})$  from Wang & Wu (1991), which requires no grouping.

As an alternative we can perform a conditional analysis by comparing  $v(\bar{y}_{st})$  and  $v_{GB}(\bar{y}_{st})$  for a given grouping of units. Assume  $n_h = 2m_h$  and the notation in (2). From the decomposition

$$\sum_i (y_{hi} - \bar{y}_h)^2 = SS_h(b) + SS_h(w),$$

where

$$SS_h(b) = m_h(z_{h1} - \bar{y}_h)^2 + m_h(z_{h2} - \bar{y}_h)^2$$

is the between-group sum of squares and

$$SS_h(w) = \sum_1 (y_{hi} - z_{h1})^2 + \sum_2 (y_{hi} - z_{h2})^2,$$

$\Sigma_1$  and  $\Sigma_2$  denoting the summation over units in the first and second groups respectively, is the within-group sum of squares, we can show that

$$v_{GB}(\bar{y}_{st}) = v(\bar{y}_{st}) + \sum_h W_h^2(n_h - 2)n_h^{-1}(n_h - 1)^{-1}\{MS_h(b) - MS_h(w)\}, \quad (3)$$

where

$$MS_h(b) = SS_h(b)/1, \quad MS_h(w) = SS_h(w)/(n_h - 2)$$

are the between-group and within-group mean squares. From (3) we have the following observations.

- (i) If  $MS_h(b) > MS_h(w)$  for all  $h$ , then  $v_{GB}(\bar{y}_{st}) > v(\bar{y}_{st})$ .
- (ii) To make  $v_{GB}$  as close as possible to  $v$ , we should group the units so that  $MS_h(b) \approx MS_h(w)$  for all  $h$ .
- (iii) It is not advisable to include units with similar  $y_{hi}$  values in the same group. If the within-group units are homogeneous,  $MS_h(w) = 0$  and the relative bias  $(v_{GB} - v)/v$  is a weighted average of  $(n_h - 2)$  with weights proportional to  $W_h^2 n_h^{-1}(n_h - 1)^{-1} MS_h(b)$ . In this extreme case,  $v_{GB}$  can be made equal to  $v$  if the  $W_h^2$  in  $v_{GB}$  is changed to  $W_h^2(n_h - 1)^{-1}$ .

### 3. RESAMPLING INFERENCE BASED ON MIXED ORTHOGONAL ARRAYS

To reduce the efficiency loss due to grouping, we can form  $R$  replicates according to a mixed orthogonal array. A mixed orthogonal array of strength  $d$ , OA  $(R, n_1 \times \dots \times n_L, d)$  is an  $R \times L$  matrix whose  $h$ th column has  $n_h$  symbols such that for any  $d$  columns each possible combination of symbols appears equally often. The  $n_h$  symbols represent the  $n_h$  units in stratum  $h$ . An example is given in Table 1. Since arrays with strength  $d \geq 3$  require a much larger  $R$  and the resulting gain in efficiency seems not worthwhile and is in general difficult to ascertain, we will only consider arrays of strength 2 and use the abbreviated notation OA  $(R, n_1 \times \dots \times n_L)$ . From this array we can form  $R$  replicates as follows. The  $r$ th replicate consists of one observation per stratum defined by the  $r$ th row

Table 1. *The first five columns form an OA (12, 3 × 2<sup>4</sup>). In the full array for 3 × 2<sup>6</sup>, only the pairs of strata 4 and 6, 5 and 7 are not orthogonal*

|   |   |   | $h$ |   |   |   |
|---|---|---|-----|---|---|---|
| 1 | 2 | 3 | 4   | 5 | 6 | 7 |
| 0 | 0 | 0 | 0   | 0 | 0 | 0 |
| 0 | 0 | 1 | 0   | 1 | 0 | 1 |
| 0 | 1 | 0 | 1   | 1 | 1 | 1 |
| 0 | 1 | 1 | 1   | 0 | 1 | 0 |
| 1 | 0 | 0 | 1   | 1 | 1 | 0 |
| 1 | 0 | 1 | 1   | 0 | 1 | 1 |
| 1 | 1 | 0 | 0   | 1 | 0 | 0 |
| 1 | 1 | 1 | 0   | 0 | 0 | 1 |
| 2 | 0 | 0 | 1   | 0 | 0 | 1 |
| 2 | 0 | 1 | 0   | 1 | 1 | 0 |
| 2 | 1 | 0 | 0   | 0 | 1 | 1 |
| 2 | 1 | 1 | 1   | 1 | 0 | 0 |

of the array. Let  $i(r, h)$  be one of  $1, \dots, n_h$  given by the  $(r, h)$ th entry of the array and  $y_{i(r, h)} = y_{hi}$  with  $i = i(r, h)$ . So the  $r$ th replicate consists of  $y_{i(r, h)}$ , for  $h = 1, \dots, L$ .

For variance estimation Gupta & Nigam (1987) proposed the estimator

$$\tilde{v}(\bar{y}_{st}) = R^{-1} \sum_r (\tilde{y}^{(r)} - \tilde{y}^{(\cdot)})^2,$$

where

$$\tilde{y}^{(r)} = \sum_h P_h y_{i(r, h)}, \quad \tilde{y}^{(\cdot)} = R^{-1} \sum_r \tilde{y}^{(r)},$$

and showed that it equals  $v(\bar{y}_{st})$ . The problem is that this method fails for nonlinear parameters. Take, for example,  $\theta = g(\bar{Y})$ , where  $\bar{Y} = \sum W_h \bar{Y}_h$  is the population mean,  $\bar{Y}_h$  is the mean of the  $h$ th stratum and  $g$  is a smooth function. Their variance estimator for  $\hat{\theta} = g(\bar{y}_{st})$  would be

$$\tilde{v}(\hat{\theta}) = R^{-1} \sum_r (\tilde{\theta}^{(r)} - \tilde{\theta}^{(\cdot)})^2,$$

where  $\tilde{\theta}^{(r)} = g(\tilde{y}^{(r)})$  and  $\tilde{\theta}^{(\cdot)} = R^{-1} \sum \tilde{\theta}^{(r)}$ . The source of the problem is the bias of  $\tilde{y}^{(r)}$  and  $\tilde{y}^{(\cdot)}$ , that is,

$$E(\tilde{y}^{(r)}) - \bar{Y} = E(\tilde{y}^{(\cdot)}) - \bar{Y} = \sum_h (P_h - W_h) \bar{Y}_h = \sum_{n_h \geq 3} W_h \{(n_h - 1)^{-1} - 1\} \bar{Y}_h,$$

which converges to zero if and only if

$$\sum_{n_h \geq 3} W_h \bar{Y}_h \rightarrow 0 \quad (4)$$

as  $n, N \rightarrow \infty$ , where  $n = n_1 + \dots + n_L$ . Denote  $\sum_h P_h \bar{Y}_h$  by  $\tilde{Y}$ . Then the bias  $\tilde{Y} - \bar{Y}$  is not asymptotically negligible if (4) does not hold. Assume that  $\max_h (n W_h)/n_h$  is bounded, which implies

$$\sum_h P_h^2 = O(n^{-1}), \quad (5)$$

$\tilde{y}^{(r)} - \tilde{Y} = O_p(n^{-1/2})$  and  $\tilde{y}^{(\cdot)} - \tilde{Y} = O_p(n^{-1/2})$ . We now have the Taylor expansions

$$\tilde{\theta}^{(r)} = g(\tilde{Y}) + (\tilde{y}^{(r)} - \tilde{Y})g'(\tilde{Y}) + o_p(n^{-1/2}),$$

$$\tilde{\theta}^{(\cdot)} = g(\tilde{Y}) + (\tilde{y}^{(\cdot)} - \tilde{Y})g'(\tilde{Y}) + o_p(n^{-1/2}),$$

from which we can show that under certain regularity conditions the leading term for  $\tilde{v}(\hat{\theta})$  is

$$R^{-1} \sum_r (\tilde{y}^{(r)} - \tilde{y}^{(\cdot)})^2 \{g'(\tilde{Y})\}^2 = v(\bar{y}_{st}) \{g'(\tilde{Y})\}^2.$$

Since the asymptotic variance of  $\hat{\theta}$  is  $\text{var}(\bar{y}_{st}) \{g'(\tilde{Y})\}^2$ , the bias of  $\tilde{v}(\hat{\theta})$  has the leading term  $v(\bar{y}_{st}) [\{g'(\tilde{Y})\}^2 - \{g'(\bar{Y})\}^2]$ . Therefore  $\tilde{v}(\hat{\theta})$  is asymptotically inconsistent except when (i)  $g$  is linear, a result proved by Gupta & Nigam (1987), or (ii) the condition (4) holds. Note that (4) is an unrealistic assumption if a nontrivial portion of strata have  $n_h \geq 3$ . This inconsistency result can be easily extended to  $\theta$  that are smooth functions of a vector of population means.

A correct method is proposed as follows. Noting the bias of  $\tilde{y}^{(r)}$ , we define, for the  $r$ th replicate,

$$\bar{y}^{(r)} = \tilde{y}^{(r)} + \sum_h (W_h - P_h) \bar{y}_h = \bar{y}_{st} + \sum_h P_h (y_{i(r,h)} - \bar{y}_h), \quad (6)$$

$$\hat{\theta}^{(r)} = g(\bar{y}^{(r)}), \quad (7)$$

and the balanced repeated replications variance estimator

$$v_B(\hat{\theta}) = R^{-1} \sum_r (\hat{\theta}^{(r)} - \hat{\theta}^{(\cdot)})^2, \quad (8)$$

where  $\hat{\theta}^{(\cdot)} = R^{-1} \sum \hat{\theta}^{(r)}$ . The scaling in (6) is called internal scaling by Wu (1986) and is used by Rao & Wu (1988) for a bootstrap method. When  $n_h = 2$ ,  $P_h = W_h$  and  $\bar{y}^{(r)} = y_{i(r,h)}$ , which imply that the formulae (7) and (8) include as a special case the standard formulae for balanced half-samples.

Let  $r_{hi}$  be the number of occurrences of symbol  $i$  in the  $h$ th stratum and  $r_{hk,ij}$  be the number of joint occurrences of symbol  $i$  in the  $h$ th stratum and symbol  $j$  in the  $k$ th stratum. Since the  $R$  replicates are selected according to a mixed orthogonal array of strength two,  $r_{hi} = R/n_h$  for each  $i$  and  $h$ , and  $r_{hk,ij} = R/(n_h n_k)$  for each  $i, j$  and  $h, k$ , which we call the first order and second order balance. From them it is easy to show the desirable properties

$$R^{-1} \sum_r \bar{y}^{(r)} = \bar{y}_{st}, \quad (9)$$

$$v_B(\bar{y}_{st}) = v(\bar{y}_{st}). \quad (10)$$

From (5), we have  $\bar{y}^{(r)} - \bar{y}_{st} = O_p(n^{-\frac{1}{2}})$  and the Taylor expansion

$$\hat{\theta}^{(r)} = g(\bar{y}^{(r)}) = g(\bar{y}_{st}) + (\bar{y}^{(r)} - \bar{y}_{st}) g'(\bar{y}_{st}) + o_p(n^{-\frac{1}{2}}). \quad (11)$$

From (10), (11) and further regularity conditions as stated in § 2.1 of Rao & Wu (1985), we can show that

$$v_B(\hat{\theta}) = v(\bar{y}_{st}) \{g'(\bar{y}_{st})\}^2 + o_p(n^{-1}). \quad (12)$$

Since the linearization variance estimator  $v(\bar{y}_{st}) \{g'(\bar{y}_{st})\}^2$  is asymptotically consistent for estimating the variance of  $\hat{\theta} = g(\bar{y}_{st})$ ,  $v_B(\hat{\theta})$  is consistent. Note that  $\tilde{v}(\hat{\theta})$  is inconsistent because  $\tilde{y}^{(r)} - \bar{y}_{st}$  and  $\tilde{y}^{(r)} - \bar{Y}$  being generally of the order  $O_p(1)$  instead of  $O_p(n^{-\frac{1}{2}})$ , do not allow a similar Taylor expansion.

Similarly the proposed method can lead to a point estimator with a lower order bias than the original estimator  $\hat{\theta}$ . Using the expansion

$$\hat{\theta}^{(r)} = \theta + (\bar{y}^{(r)} - \bar{Y}) g'(\bar{Y}) + \frac{1}{2} (\bar{y}^{(r)} - \bar{Y})^2 g''(\bar{Y}) + o_p(n^{-1}),$$

and a similar one for  $\hat{\theta}$ , we have, up to lower order terms

$$\begin{aligned} \text{bias}(\hat{\theta}^{(r)}) &= \text{bias}(\hat{\theta}^{(\cdot)}) = \frac{1}{2} g''(\bar{Y}) E(R^{-1} \sum_r (\bar{y}^{(r)} - \bar{Y})^2) \\ &= g''(\bar{Y}) \text{var}(\bar{y}_{st}) = 2 \text{bias}(\hat{\theta}). \end{aligned}$$

Therefore the estimator

$$\hat{\theta}_B = 2\hat{\theta} - \hat{\theta}^{(\cdot)} \quad (13)$$

has a lower order bias than  $\hat{\theta}$ . A special case of  $\hat{\theta}_B$  for  $n_h = 2$  was proposed in formula (20) of Rao & Wu (1985). The stated result can be proved by using an argument similar to that in their § 2.2.

The previous results for the proposed method hold for  $\theta = g(\bar{Y}_1, \dots, \bar{Y}_k)$ , where  $\theta$  is a smooth function of a vector of means. This class includes several parameters of interest such as the ratio, correlation and regression coefficients. The technical derivations are similar to those in Rao & Wu (1985). Finally, for simple random sampling without replacement within each stratum, the proposed method and the stated results have a straightforward extension: replace  $P_h^2 = W_h^2(n_h - 1)^{-1}$  by  $W_h^2(1 - f_h)(n_h - 1)^{-1}$ , where  $f_h = n_h/N_h$  is the sampling fraction in stratum  $h$ .

#### 4. EXTENSIONS TO GENERAL STATISTICS

The method proposed in § 3 can be extended to more general  $\theta = T(F)$ , where  $F = \sum_h W_h F_h$  with  $F_h = N_h^{-1} \sum_i \delta_{y_{hi}}$  being the population distribution function of the  $h$ th stratum,  $\delta_{y_{hi}}(a) = I(y_{hi} \leq a)$ , and  $T$  is a smooth functional of  $F$ . Typically  $\theta$  is estimated by  $\hat{\theta} = T(\hat{F})$ , where  $\hat{F} = \sum W_h \hat{F}_h$  with  $\hat{F}_h$  being the sample analog of  $F_h$ . We assume that  $T$  admits the expansion

$$\begin{aligned} T(\hat{F}) &= T(F) + T(F; \hat{F} - F) + o_p(n^{-\frac{1}{2}}) \\ &= T(F) + \sum_{h=1}^L W_h n_h^{-1} \sum_{i=1}^{n_h} T(F; \delta_{y_{hi}} - F_h) + o_p(n^{-\frac{1}{2}}), \end{aligned} \quad (14)$$

where

$$T(F; G - F) = \left[ \frac{d}{dt} T\{F + t(G - F)\} \right]_{t=0}$$

is the Gâteaux derivative of  $T$ . For  $\theta = F^{-1}(p)$ , the  $p$ -quantile of the population,  $T(F; \delta_y - F_h) = F_h(\theta) - \delta_y(\theta)$ , and (14) is established in an unpublished manuscript by J. Shao and C. F. J. Wu. Other examples of  $T$  are given by Serfling (1980, Ch. 6).

As extensions of (6) and (7), we define

$$\hat{F}^{(r)} = \hat{F} + \sum_h P_h (\hat{F}_h^{(r)} - \hat{F}_h), \quad (15)$$

$$\hat{\theta}^{(r)} = T(\hat{F}^{(r)}), \quad (16)$$

where  $\hat{F}_h^{(r)}(a) = I(y_{i(r,h)} \leq a)$  and the variance estimator  $v_B(\hat{\theta})$  as in (8). For the sample  $p$ -quantile  $\hat{\theta} = \hat{F}^{-1}(p)$ , the asymptotic consistency of  $v_B(\hat{\theta})$  can be established by using (14), the orthogonality of the array and some complicated probabilistic arguments along the lines of Shao & Wu. For general  $\hat{\theta}$  satisfying (14), the proof of consistency of  $v_B(\hat{\theta})$  is technically more cumbersome and difficult.

#### 5. USES OF NEARLY ORTHOGONAL ARRAYS

The proposed methods in §§ 3 and 4 hinge on the availability of mixed orthogonal arrays for the given sample sizes  $n_h$  ( $h = 1, \dots, L$ ). Large collections of such arrays are given by Dey (1985), Wang & Wu (1991) and the Madison Ph.D. thesis of J. C. Wang.

Often the orthogonality assumption imposes a severe requirement on  $R$ , the number of replicates. For example, for the  $\text{OA}(12, 3 \times 2^m)$  in Table 1, the maximum  $m$  is 4. For  $m \geq 5$ ,  $R$  in  $\text{OA}(R, 3 \times 2^m)$  has to go up to 24 from 12. Since it is impossible to find an economical mixed orthogonal array for each possible  $n_h$  ( $h = 1, \dots, L$ ) we may group the units as in § 2 and then use a more economical array for the grouped units. The grouping can be done according to the rules in § 2. Suppose that  $n_1 = n_2 = 3$ ,  $n_3 = n_4 = n_5 = 2$ . The smallest  $\text{OA}(R, 3^2 \times 2^3)$  has  $R = 36$ . By grouping the 3 units in the second stratum to 2 units, we need only 12 replicates by using the  $\text{OA}(12, 3 \times 2^4)$  in Table 1.

An alternative method for saving replicates is to employ arrays in which most, but not all, pairs of columns are orthogonal. Consider the case of  $n_1 = 3$  and  $n_h = 2$  for  $h = 2, \dots, 7$ . A mixed orthogonal array would require 24 replicates. Using the full array for  $3 \times 2^6$  in Table 1 only requires 12 replicates. Then the question is: how much efficiency loss is caused by the nonorthogonal pairs of columns?

Let  $r_{hi}$  and  $r_{hk,ij}$  be as defined before (9). Then nonorthogonality can be measured by the differences

$$\Delta_{hi} = r_{hi}R^{-1} - n_h^{-1}, \quad \Delta_{hk,ij} = r_{hk,ij}R^{-1} - n_h^{-1}n_k^{-1}.$$

Let  $\bar{y}^{(r)}$ ,  $\hat{\theta}^{(r)}$ ,  $v_B(\bar{y}_{st})$  and  $v_B(\hat{\theta})$  be defined as in § 3. Then it can be shown that

$$R^{-1} \sum_r \bar{y}^{(r)} - \bar{y}_{st} = \sum_h P_h \sum_i \Delta_{hi} (y_{hi} - \bar{y}_h), \quad (17)$$

$$v_B(\bar{y}_{st}) - v(\bar{y}_{st}) = \sum_h P_h^2 \sum_i \Delta_{hi} (y_{hi} - \bar{y}_h)^2 + \sum_{h \neq k} P_h P_k \sum_{i,j} \Delta_{hk,ij} (y_{hi} - \bar{y}_h)(y_{kj} - \bar{y}_k). \quad (18)$$

In spite of (17),  $\bar{y}^{(r)} - \bar{y}_{st}$  ( $r = 1, \dots, R$ ) have  $O_p(n^{-1/2})$  under the assumption for (5), which allow  $\hat{\theta}^{(r)}$  to be expanded around  $\hat{\theta}$  as in § 3. Using derivations similar to (12), we have  $v_B(\hat{\theta}) \doteq v_B(\bar{y}_{st})\{g'(\bar{y}_{st})\}^2$ . It becomes clear that the bias and inconsistency of  $v_B(\hat{\theta})$  is due to (18), not (17). From (18) the bias of  $v_B$  due to nonzero  $\Delta_{hi}$  and  $\Delta_{hk,ij}$  can be made small if few of the  $\Delta$ 's are nonzero and large  $|\Delta_{hi}|$  and  $|\Delta_{hk,ij}|$  can be assigned to strata with small  $P_h$ ,  $P_k$ ,  $s_h^2 = (n_h - 1)^{-1} \sum_i (y_{hi} - \bar{y}_h)^2$ ,  $s_k^2$  and  $s_{hk}$ .

The nearly orthogonal array for  $3 \times 2^6$  in Table 1 is a good example since  $\Delta_{hi} = 0$  for all  $h, i$  and the only nonzero  $\Delta_{hk,ij}$  are  $\Delta_{46,ij}$  and  $\Delta_{57,ij}$ , which take  $\pm \frac{1}{12}$ . A collection of arrays with  $\Delta_{hi} = 0$  and few nonzero  $\Delta_{hk,ij}$  is given in an unpublished manuscript by J. C. Wang and C. F. J. Wu.

Another class of arrays with proportional frequencies, that is  $r_{hk,ij} = R^{-1}r_{hi}r_{kj}$ , was discussed by Gupta & Nigam (1987). They can be obtained from orthogonal arrays through collapsing factors. For example, by assigning the levels of the first factor of  $\text{OA}(9, 3^4)$  according to  $0 \rightarrow 0$ ,  $1 \rightarrow 1$  and  $2 \rightarrow 1$ , we obtain an array of 9 runs for  $2 \times 3^3$ . For any collapsed factor  $h$ , both  $\Delta_{hi}$  and  $\Delta_{hk,ij}$  are nonzero for any  $h \neq k$ . In this case,  $\Delta_{1i} = \pm \frac{1}{6}$  and  $\Delta_{1k,ij} = \pm \frac{1}{18}$  for  $k = 2, 3, 4$ . Since collapsing factors causes a substantial number of  $\Delta$ 's to be nonzero, from (18) the bias of  $v_B(\bar{y}_{st})$  will be nonnegligible. In general this class of arrays is not recommended.

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